AOE3

AGENT OPERATING ENGINE

Deterministic Authorization, Execution Control, and Verifiable Evidence for Autonomous AI Systems

INTELLIGENCE IS NOT AUTHORIZATION.

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Clarified design edition — REVIEW CANDIDATE; not a newly authorized release

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Reader’s guide

This edition prioritizes legibility and traceability over decorative styling. Diagrams show intended logical boundaries, not evidence of deployed non-bypassability. The original 32-section structure is preserved in substance and sequence.

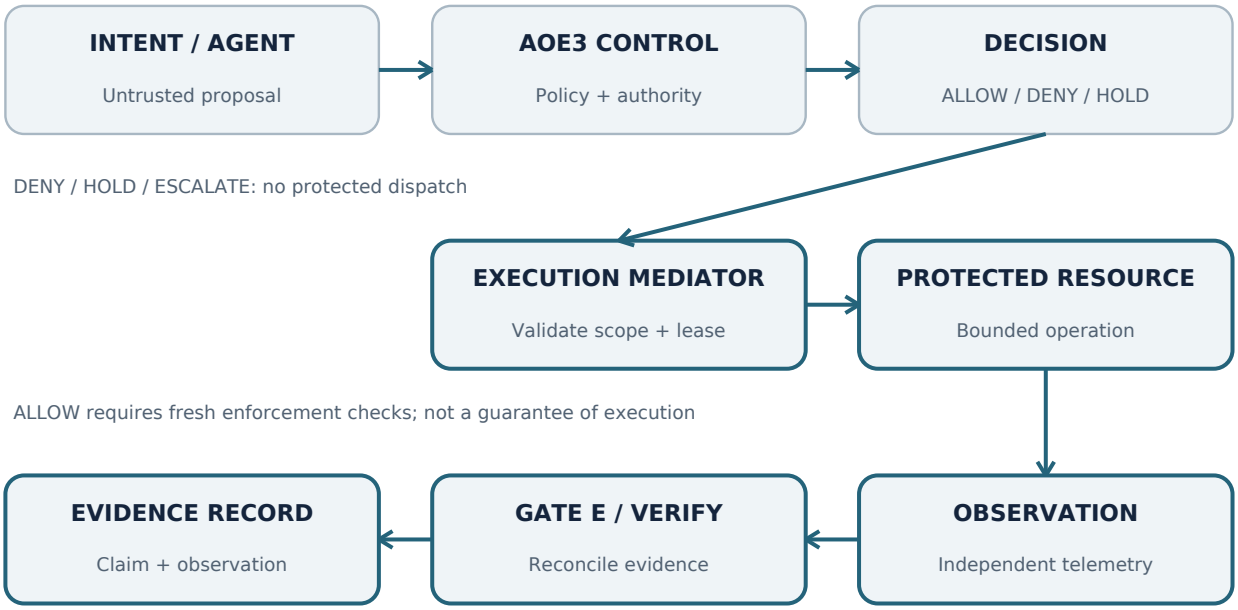
Interpretation rules

PROPOSAL ≠ AUTHORIZATION · ALLOW ≠ EXECUTED · SIGNED ≠ OBSERVED · HASHED ≠ TRUE · DESIGN ≠ QUALIFIED

Diagram legend

Boxes represent distinct responsibilities. Arrows indicate intended information or control flow, not automatically verified deployment paths. DENY, HOLD and ESCALATE never imply dispatch.

01 / Consequential-action lifecycle



The executor must validate the exact authorized target, operation, parameters, context and lease at the dispatch boundary. An ALLOW decision alone is insufficient.

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01 The Problem

Agentic systems can call APIs, write to databases, change business records, trigger financial or operational processes, and coordinate multi-step workflows. IAM, RBAC/ABAC, API gateways, monitoring, and human approval remain essential but may not resolve whether this exact action is authorized now, against this target, with this evidence and context. IS THIS EXACT ACTION AUTHORIZED NOW?

02 Core Principle

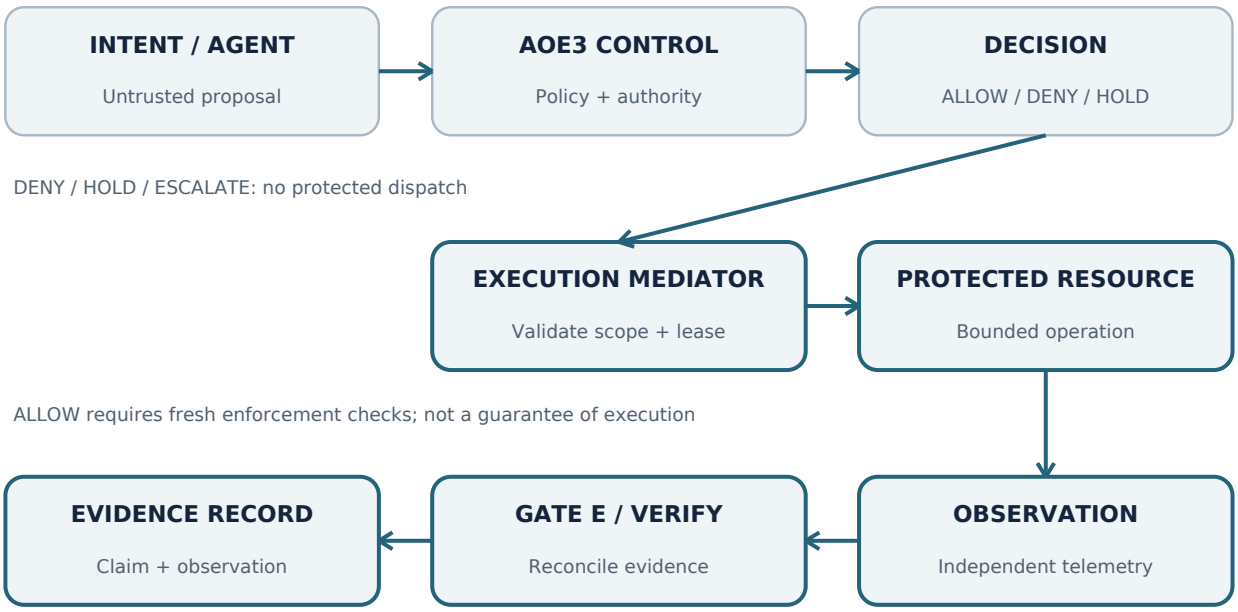
Intelligence and authority are distinct system properties. A model may reason correctly, hold credentials, and generate a technically executable request. None of these facts independently establishes permission. The proposing component must be structurally separate from the mechanism determining authority. AI PROPOSAL IS NOT AUTHORIZATION.

03 Capability vs Authority

Capability is what a system can technically perform. Authority is what it is permitted to perform under defined policy, evidence, context, time, scope, and governance. A system can possess capability without authority. CAN EXECUTE IS NOT MAY EXECUTE.

04 Architectural Model

The consequential action lifecycle separates user/system intent, agent proposal, deterministic control, mediated execution, protected resource, independent observation, reconciliation, and evidence. Decisions may include ALLOW, DENY, ESCALATE, or HOLD. CONSEQUENTIAL ACTION LIFECYCLE



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05 AOE3 Control

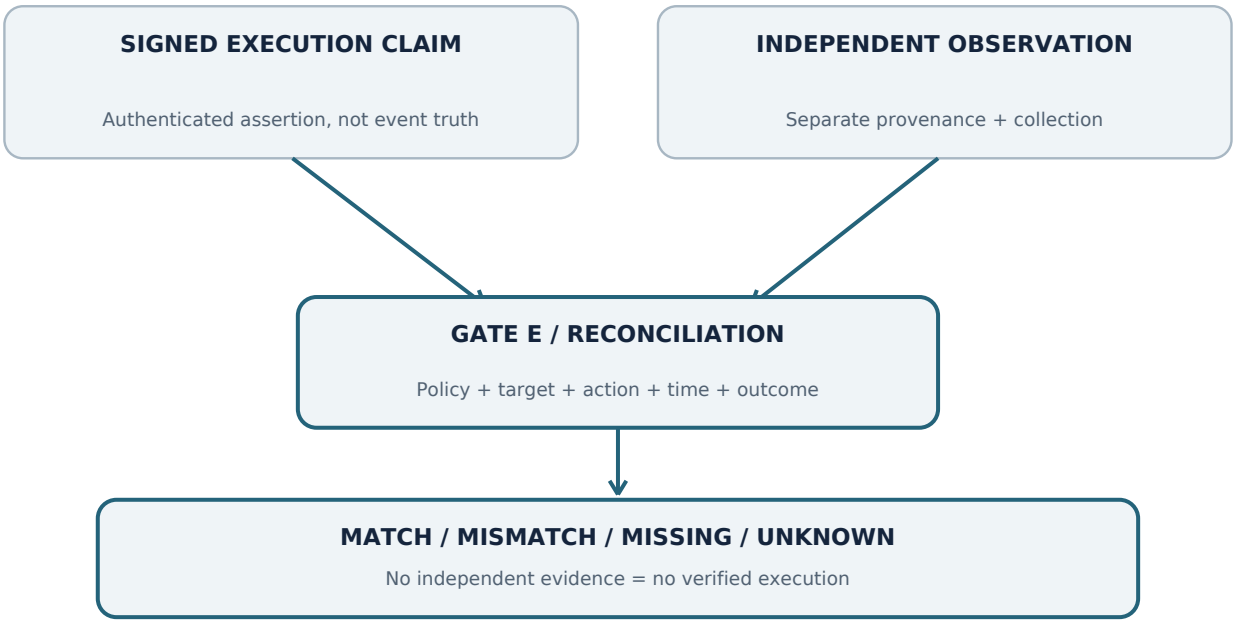
AOE3 Control evaluates action identity, agent identity, target, permitted operation, policy version, evidence presence and integrity, transaction bounds, context, authorization lifetime, human approval, replay status, and environmental constraints. ALLOW satisfies required conditions; DENY fails a condition; ESCALATE requests additional authority; HOLD indicates insufficient established authority. Missing evidence cannot silently become permission. UNKNOWN IS NOT PASS.

06 Execution Boundary

Authorization does not prove correct execution or universal non-bypassability. A deployment must define its controlled executor, credential scope, target, operation, parameters, validity period, replay controls, failure behavior, alternate paths, and bypass assumptions. Claims of control are bounded to the paths actually mediated. PERMIT IS NOT EXECUTION.

07 Independent Verification — Gate E

Cryptographic authenticity and execution truth are distinct. A signed record authenticates a claim within its cryptographic context; it does not prove the event happened as described. Gate E compares authorized claims with independently observed execution evidence and reports MATCH, MISMATCH, MISSING, or UNKNOWN. Required evidence cannot be invented. SIGNED CLAIM IS NOT EXECUTION TRUTH.



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08 Evidence Architecture

Evidence is a first-class output. A defensible package may bind proposal identity, policy decision, policy version, authorization, action parameters, target, context, timestamps, inventory, lifetime, replay state, observation, verification, provenance, digests, and signature state. Independent reviewers should not need to trust the model's own explanation.

09 Cryptographic Integrity

Hashes identify exact bytes; signatures may authenticate a signed claim. Cryptography supports byte identity, origin authentication, tamper detection, artifact linkage, and provenance. Neither a hash nor a signature independently proves an external event or legal validity. HASHED DOES NOT MEAN TRUE.

10 Human Authority

AOE3 preserves human decision-making. A policy decision may allow, deny, or escalate. Escalation can require a human approval, denial, or expiry event. Human decisions become evidence and remain subject to validity, scope, and expiration.

11 Fail-Closed Operation

Consequential operations must deny or hold when required evidence is missing or invalid, authorization expires, a token is replayed, policy or target mismatches, parameters diverge, verification fails, observations conflict, or authority remains unresolved. Indeterminate states are not converted into approval. UNKNOWN IS NOT PASS.

12 Model and Provider Independence

AOE3 is intended to work conceptually with local or cloud models, different providers, agent frameworks, and enterprise systems. Intelligence remains with the model; authorization and verification remain independently governed. Integration does not transfer architectural authority.

13 Sovereign Profile

AOE3 Sovereign is a higher-assurance operating profile, not a replacement for the core architecture. It may add local execution, network containment, residency, key custody, restricted administration, isolated deployment, retention, and independent verification. Exact requirements depend on the deployment.

14 Product Layers

AOE3 GovControl focuses on consequential-action authorization, policy, and operational control. AOE3 Assurance focuses on evidence, verification, conformance, and auditability. AOE3 Sovereign focuses on stronger isolation, custody, sovereignty, and deployment constraints. All share the same architectural doctrine.

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15 Demonstration vs Production

A demonstrator may show deterministic decisions, fail-closed behavior, evidence generation, replay handling, expiry, and verification logic. It does not establish production qualification. Each environment requires separately scoped implementation, empirical evidence, and qualification. DEMONSTRATED IS NOT PRODUCTION QUALIFIED.

16 Claim Discipline

Each claim should identify exact wording, evidence state, artifact, scope, limitations, version, and authorized audience. The chain connects claim to product, component, version, source artifact, evidence, verification scope, qualification state, and disclosure authority. External claims must not exceed evidence.

17 Evidence-State Distinctions

DECLARED means asserted; IMPLEMENTED means present in code; EXECUTED means run in a defined environment; OBSERVED means execution evidence captured; VERIFIED means checked against criteria; QUALIFIED means conditions satisfied; PROMOTED means explicitly elevated by authorized decision. These states cannot be silently collapsed.

18 Security Boundary

The intended controls include least privilege, deterministic policies, evidence requirements, short-lived authority, replay resistance, separated credentials, bounded executors, network control, independent observation, fail-closed decisions, and auditability. Actual enforcement remains implementation- and deployment-dependent.

19 Enterprise Integration

AOE3 is intended to complement IAM, RBAC/ABAC, gateways, SIEM, policy engines, workflow systems, clouds, databases, enterprise applications, and agent frameworks. Its distinct proposition is an independent consequential-action boundary between proposal and protected execution.

20 Example Use Cases

Financial operations: evaluate an agent-proposed payment against amount, authority, policy, and context. Government services: separate recommendation from public-service authority. Enterprise administration: mediate proposed record changes. Critical infrastructure: govern operational commands and independently reconcile outcomes. These are potential categories, not claims of qualified deployments.

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21 Commercial Model

AOE3 is positioned as a product and private-intellectual-property company. Bounded professional or assessment services can support discovery, readiness, evidence development, integration design, and controlled pilot preparation without redefining the corporate identity as a consultancy. SERVICE DELIVERY IS NOT CORPORATE IDENTITY.

22 Ownership and Authority

Architecture, product direction, evidence promotion, disclosure, release, and governance remain subject to owner authority. Infrastructure provision, review, pilot participation, API access, credits, NDAs, and commercial discussions do not independently confer architectural authority. Partnership and governance authority are separate.

23 Source-of-Truth Discipline

Evidence hierarchy: (1) owner-authorized canonical signed or sealed artifacts and verifiable receipts; (2) canonical manifests and independently matched digests; (3) approved registers and owner directives; (4) drafts and analytical reviews; (5) recollections and unverified claims. Artifact integrity is not substantive correctness.

24 Current Qualification Position

This document states architecture and product philosophy. It is not evidence of production qualification, completed provider integrations, universal non-bypassability, regulatory certification, empirical demonstration of every control, or commercial partnership. Each such claim requires independent supporting evidence.

25 Strategic Principle

AI may propose. Authority must be independently established. Execution must remain controlled. What actually happened must be independently verifiable. CONTROL EXECUTION. VERIFY EVIDENCE.

26 Conclusion

Autonomous AI creates substantial capability but should not thereby acquire consequential authority. AOE3 separates proposal from authorization, authorization from execution, execution from observation, observation from verification, and verification from promotion. This is a framework for powerful AI under accountable human and institutional authority.

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27 Technical Architecture: Separation of Planes

The reference design separates the reasoning engine from protected state-changing interfaces. The Reasoning Plane generates unsigned structured intent. The Authorization Plane evaluates cryptographic identity, policy, scope, replay, and time-bound authorization. The Execution Plane mediates protected operations within the authorized scope. 01 REASONING PLANE Probabilistic intelligence / unsigned proposal 02 AUTHORIZATION PLANE Deterministic policy / scoped authorization 03 EXECUTION PLANE Mandatory mediation / protected state



28 Cryptographic Source Identity and Authentication

Action provenance can be integrity-bound and authenticated under an applicable cryptographic profile. Ed25519 is used in defined evidence/signature contexts; key custody and lifecycle are separate obligations. The Key 60 design requirement defines a 60-second authorization-lease validity window. Lease expiry does not prove key destruction or eliminate key compromise. LEASE EXPIRY IS NOT KEY DESTRUCTION.

Clarification: a 60-second lease validity window does not imply that the signing key is destroyed at expiry. Key destruction requires its own independently evidenced lifecycle.

29 Non-Compensatory Containment Mechanics

Non-compensatory containment prevents a secondary privilege or heuristic override from compensating for a failed primary gate. Semantic checks, signatures, lease expiry, and rate limits are independent constraints. If a required check fails or becomes indeterminate, protected execution is denied and telemetry is emitted.

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30 Network Boundary Semantics

Declarative CNI policy expresses intended connectivity, not proof of enforcement. Actual behavior depends on datapath, implementation, propagation, and deployment. eBPF cgroup/connect and socket hooks can mediate connections; payload-aware controls require an application-semantic enforcement point. Tracepoints observe; mandatory controls such as LSM, seccomp, sandboxing, or explicit mediators enforce. gVisor can reduce host-kernel syscall exposure but does not natively validate AOE3 leases.

Control distinction: network socket hooks can restrict connections; application-semantic authorization requires mediation capable of inspecting the relevant action. Tracepoints observe; LSM, seccomp or explicit execution mediators can enforce within their configured scope.

31 Independent Verification: Execution Reconciliation

Gate E independently reconciles an authorized claim, controlled execution, signed claim/evidence record, and external telemetry. The oracle reports MATCH, MISMATCH, MISSING, or UNKNOWN. A signed claim alone never establishes execution truth; absent independent evidence must remain absent.

A MATCH result requires independently sourced observations matched to the authorized action, target, operation, timing and outcome. A signed claim alone cannot establish MATCH.

32 Technical Conclusion

AOE3 combines proposed datapath enforcement, cryptographic source identity, non-compensatory containment, and independent reconciliation to reduce dependence on probabilistic self-policing. This architectural formalization aligns with AOE3-STD-001 governance principles. Production effectiveness and higher assurance remain subject to implementation, evidence, and qualification.

Correction and qualification register

The following items remain deployment-specific verification obligations, not proven results.

1. Establish complete inventory of alternative execution paths and demonstrate that protected operations cannot bypass the selected mediator.

2. Verify fail-closed handling for expired leases, replay, missing evidence, indeterminate policy results and inconsistent observations.

3. Demonstrate independent telemetry provenance and time correlation, including clock uncertainty and evidence gaps.

4. Document key custody, rotation, revocation, and destruction separately from authorization-lease expiry.

5. Validate actual eBPF/LSM/seccomp/gVisor attachment points and their enforced semantics in the target environment.

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6. Maintain explicit distinction between declared architecture, implemented controls, executed tests, observed evidence and qualified deployment.

Rights and status

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